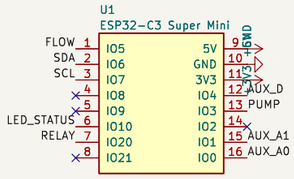
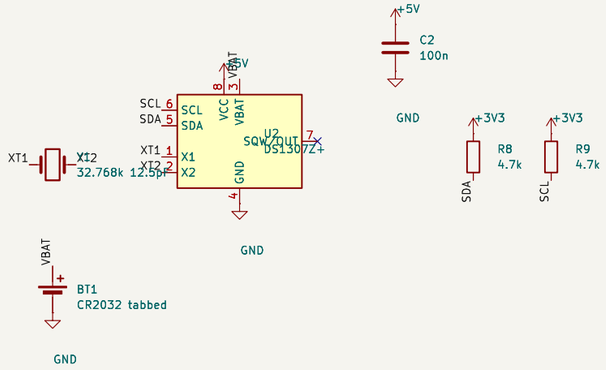
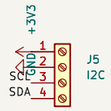
MCU: ESP32-C3 Super Mini, plugged into 2 x 8-pin sockets.
Its USB-C powers the board (5 V / 1 A charger in the field): the 5V pin
(USB VBUS) feeds the pump, the relay coil, the flow sensor and the RTC;
the 3V3 pin (on-module LDO) feeds the sensors and pull-ups.
Left free: GPIO2 and GPIO8/9 (strapping; GPIO8 = module LED, GPIO9 = BOOT), GPIO21.



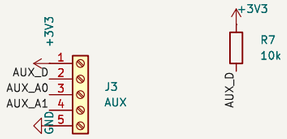

RTC: DS1307 at 0x68, on 5 V (it needs 4.5-5.5 V).
I2C pull-ups go to 3V3: the DS1307 reads 3.3 V as high (VIH 2.2 V).
Backup: CR2032 with solder tabs, standing vertical (no holder).
Crystal 32.768 kHz, 12.5 pF load.



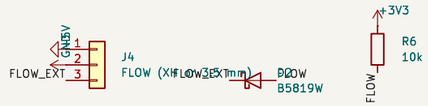

I2C terminal: 3V3 GND SCL SDA, the pin order of the ASC-AHT20
board and of most AHT20 / BMP280 / BH1750 modules (VIN GND SCL SDA).

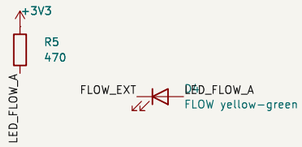



AUX: D = GPIO4 with a 10k pull-up (DHT11/DHT22 or a DS18B20 probe),
A0/A1 = GPIO0/1 on ADC1 (soil-moisture probe, LDR, pot, switch).

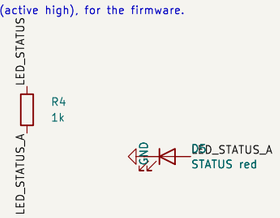



Flow: YF-S401 on 5 V (needs 5-24 V), 5880 pulses per litre.
Its output can swing to 5 V, so Schottky D2 only pulls GPIO5 low
and R6 pulls it up to 3V3. The FLOW LED sinks into the sensor
output, so it blinks with every pulse (no GPIO needed).



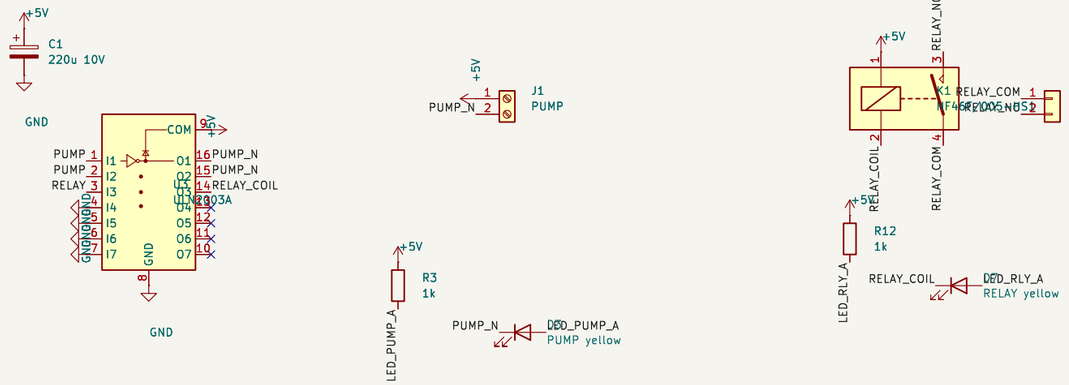




STATUS LED on GPIO10 (active high), for the firmware.




Pump: 5 V DC submersible (about 0.2-0.3 A) on ULN2003 channels 1+2 in parallel
(about 0.9 V drop, so ~4 V at the pump); the ULN2003's COM pin to +5V is the flyback clamp.
PUMP LED sits across the switch side: lit while the pump is driven.
C1 rides out the start-up surge of the pump and the relay on USB 5 V.

Relay (FG1): HF46F 5 V coil, 1 Form A, 5 A 250 VAC, for a pump contactor coil.
GPIO20 (UART0 RX), an input at boot, so the relay stays off) drives ULN2003 channel 3;
unused inputs 4-7 are grounded. RELAY LED across the coil. Contacts on a pluggable 5.08 mm
MSTBVA header; 230 V copper keeps >= 5 mm from everything else (.kicad_dru rule).




Mounting: 3 x 4.2 mm holes for 5 mm nylon reverse-mount snap standoffs, bases taped to the box floor (any box).







48 x 44 mm; 4 Minis + 4 AHT20 sticks per 100 x 100 panel; 3 snap-standoff holes (any box)
Generated by tools/gen_sch.py from tools/design.py - edit design.py and regenerate

**Agri Sensors and Controls**

Sheet: /
File: ASC-MiniC3.kicad_sch

***Title: ASC Student Kit - Mini-C3 rev B (low cost, FG1 common)***

| Size: A3 | Date: 2026-10-10 | **Rev: B** |
|---|---|---|
| KiCad E.D.A. 9.0.7 | | Id: 1/1 |